

1 - Weighted Logistic Regression

In this question, we explore the weighted version of Logistic Regression, where the training samples are weighted based on a test sample. The cost function is defined as follows:

$$J(\theta) = -\frac{\lambda}{2} \theta^T \theta + \sum_{i=1}^m w^{(i)} [t^{(i)} \log y(x^{(i)}) + (1 - t^{(i)}) \log (1 - y(x^{(i)}))]$$

- Term $-\frac{\lambda}{2} \theta^T \theta$ is the regularization term, which helps improve the performance of Newton's method.
- Set the value of the lambda parameter to 0.0001.

Tasks

1 – Closed Form Gradient and Hessian

Find the gradient vector and Hessian matrix of the cost function in closed form.

2 - Implementations

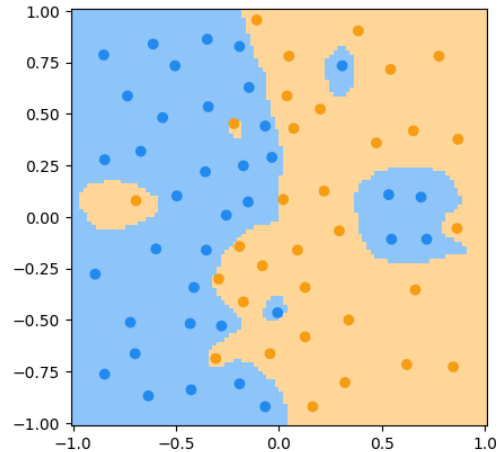
Solve the following tasks using the provided training dataset. The dataset consists of two files: features.txt and labels.txt. Each line in the respective file contains the feature vector and label of a training sample.

a - Implement simple logistic regression from scratch on the dataset. Then, display the decision regions.

b - Implement weighted logistic regression from scratch on the dataset. Then, display the resulting hypothesis. When evaluating the function, use the following equation to adjust the weights:

$$w^{(i)} = \exp\left(-\frac{(x - x^{(i)})^2}{2 \tau^2}\right)$$

Set the value of τ to 0.8.



Example of decision regions.

c - In the second part, experiment with different values of τ : 0.01, 0.05, 0.1, 0.5, 1, and 5. Then, explain how the changes in τ affect the resulting hypotheses.

2 - Predicting Cancer Tumors with Naïve Bayes Classifier

In this assignment, we will apply machine learning techniques to the Wisconsin Breast Cancer Dataset, which contains features related to breast cancer and a binary outcome (malignant or benign). Breast cancer is one of the leading causes of cancer-related deaths, and early detection significantly improves treatment outcomes. By building and evaluating predictive models, we can help identify patterns that aid in early diagnosis, ultimately contributing to better patient care.

This task will help you understand key concepts such as model performance, data splits, and the curse of dimensionality, all of which are crucial for developing reliable models in healthcare and other domains.

Tasks:

1 - Data Preprocessing and Model Training

- Split the dataset into training and test sets.
- Implement a Naïve Bayes classifier using scikit-learn to predict if a tumor is malignant or benign.
- Preprocess the data (handle missing values, scale features if needed).
- Report the model's accuracy, precision, recall, and F1-score.

2 - Curse of Dimensionality

- Define the "Curse of Dimensionality" and its impact on machine learning.
- Does it affect the Wisconsin Breast Cancer dataset? Justify your answer.
- How does it affect models with higher-dimensional feature sets, and what techniques (e.g., dimensionality reduction) can help mitigate it?

3 - Bonus: Implement Naïve Bayes from Scratch

- Implement Naïve Bayes from scratch in Python (without scikit-learn).
- Compare its performance with the scikit-learn Naïve Bayes classifier (accuracy and other metrics).
- Discuss any differences between the two implementations.

3 - Model Comparison on Pima Indian Diabetes Dataset

You are given the Pima Indian Diabetes Dataset, which includes features related to diabetes and a binary outcome (1 for diabetes, 0 for no diabetes). This dataset is important for predicting diabetes, a major health issue, and can help in early detection and intervention.

This task will focus on applying machine learning techniques to assess model performance and data handling.

Tasks:

1 - Implantation

Implement the following machine learning models using from scrach or from relevant libraries such as scikit-learn:

- Logistic Regression
- Support Vector Machine (SVM)
- XGBoost
- Decision Tree
- Naive Bayes

For each model:

Perform necessary data preprocessing. Split the dataset into training and test sets. Train each model on the training data and evaluate it on the test data. Compare the models using relevant performance metrics such as:

- Accuracy

- Precision
- Recall
- F1-score
- Matthews Correlation Coefficient (MCC)

Implement the performance metric functions on your own and explain the pros and cons of each metric.

2 - Analysis:

- After obtaining the results, compare the performance of the models.
- Discuss why some models may perform better or worse than others.
- Consider which model would be more suitable for the problem based on your analysis.

4 - Exploring SVM Parameters for Classification

Use the Pima Indian Diabetes dataset. Implement a Support Vector Machine (SVM) classifier to predict whether a person has diabetes or not.

Tasks:

1 - Train an SVM Classifier:

- Apply the SVM algorithm to the dataset.
- Split the data into training and test sets.
- Use scikit-learn or any relevant library for the implementation.

2 - Experiment with SVM Parameters:

Modify the following SVM hyperparameters:

- Kernel
- C
- Degree

3 - Explain the Effect of Parameters:

- Explain the role of each parameter (kernel, C, gamma, degree) in the SVM algorithm.
- Discuss how changes in these parameters affect the model's performance, based on metrics like accuracy, precision, recall, and F1-score.
- Use grid search to find the best set of parameters.

5 - Grade Prediction

We have data about students in a machine learning class, with two features:

- X_1 : Total hours spent studying machine learning.
- X_2 : Average grade of students before attending the course (out of 100).

We aim to use logistic regression to predict the probability of a student achieving a grade of 100 in the course. After training the model, the obtained coefficients are:

- $B_0 = 0.3$
- $B_1 = 0.005$
- $B_2 = 0.01$

Tasks:

1. Probability Calculation

Given:

- A student studied for 80 hours and has an average grade of 75.

Calculate the probability that this student will achieve a grade of 100 using the logistic regression model.

2. Hours of Study for Desired Probability

Given:

- A student has an average grade of 60.
- The student wants to achieve a 90% chance of getting a grade of 100.

Calculate the number of hours the student should study to achieve this probability.

6. Solving Bishop's Book Problems

Solve the following problems from **Bishop's book**:

- **Problem 4.9**
- **Problem 4.10**
- **Problem 4.20**

